AccessWatch: Multimodal Fusion for Hierarchy-Aware Forecasting of Humanitarian Access Risk in Ukraine

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Abstract

This paper presents **AccessWatch**, a multimodal direct forecasting framework for week-ahead humanitarian access risk modeling in Ukraine at the raion level. The system integrates ACLED-derived conflict histories, NASA FIRMS thermal anomaly features, VIIRS Black Marble nighttime-light dynamics, population and transport exposure indicators, border-access covariates, and pre-2024 UNOSAT damage summaries into a unified raion-week panel. The released experiment package contains five direct forecast horizons (week+1 through week+5), two feature-set variants (67 and 77 features), five algorithms, and five missing-modality settings, yielding 250 held-out test configurations. The final evaluation uses 139 raions and 104 weekly anchors with strict chronological train/ validation/ test splits. The strongest balanced full multimodal configuration is Model 2 with CatBoost, which achieves 0.8256 any-event F1, 0.8643 high-intensity F1, and 0.7252 event type macro-F1 on the week+1 test set while maintaining low count-regression error. An ablation study shows that autoregressive conflict-history features provide the largest consistent gain, whereas the marginal value of nighttime-light and FIRMS inputs depends on the algorithm and horizon. The resulting framework offers a technically grounded and operationally relevant prototype for humanitarian early warning under incomplete multimodal availability.

1 Introduction

Humanitarian access conditions in active conflict zones can worsen rapidly, often before those changes are clearly visible in official reports or operational updates. In practice, access risk does not arise from a single cause. It is shaped by the combined effects of conflict activity, infrastructure disruption, geographic exposure, and broader signals of instability. As a result, systems that depend on only one type of input may miss important warning signs or fail to capture how these factors interact.

AccessWatch was developed to address this challenge through a multimodal forecasting framework for Ukraine. The system combines conflict-event histories, satellite-derived indicators, and static geospatial context to generate direct week-ahead predictions of humanitarian access risk at the raion level. Rather than focusing on a single outcome, the framework is designed to support a broader view of local risk by forecasting general event occurrence, high-intensity escalation, event-type patterns, and count-based outcomes.

The forecasting task is formulated at the raion-week level. Each instance represents the state of a Ukrainian raion at a given weekly anchor point, and the model predicts outcomes for a future target week at horizon $h \in \{1, 2, 3, 4, 5\}$. This direct forecasting design avoids the error accumulation that can arise in recursive forecasting and allows separate models to be optimized for each forecast horizon. The framework also adopts a hierarchical multitask structure to ensure that coarse- and fine-grained predictions remain logically consistent.

By combining multimodal inputs with strict temporal evaluation, AccessWatch aims to provide a practical and technically grounded prototype for short-term humanitarian early warning in conflict-affected settings.

2 Motivation

AccessWatch is motivated by both an operational need and a methodological gap. From an operational perspective, humanitarian organizations benefit from short-term early warning signals that can identify districts at risk of worsening disruption before conditions are fully reflected in formal reporting. In conflict settings, access risk is shaped by multiple interacting factors, including conflict intensity, infrastructure damage, geographic exposure, and broader signs of instability. A forecasting system that relies on only one of these signals may overlook important patterns that emerge when they are considered together.

From a methodological perspective, many existing forecasting approaches have important limitations. Some rely on randomly shuffled evaluation splits that do not reflect real deployment conditions, while others simplify the task to a single binary outcome or ignore the reality that some data sources may be missing or unreliable. AccessWatch was designed to address these issues through three core principles: strict chronological evaluation, a richer hierarchy of forecasting targets beyond simple event occurrence, and explicit testing under missing-modality settings.

A further motivation was practicality under hackathon constraints. Rather than building a highly complex end-to-end remote-sensing system, this project focuses on a disciplined multimodal panel-construction pipeline combined with several families of predictive models. This makes the framework feasible to implement, easier to interpret, and sufficiently robust to support meaningful analysis and conference-style reporting.

3 Problem Statement

Given the multimodal state of a Ukrainian raion at week t, the goal of AccessWatch is to forecast whether that raion will experience signs of worsening conflict-related disruption in a future target week located exactly h weeks ahead, where $h \in \{1, 2, 3, 4, 5\}$. Rather than treating this as a single prediction task, the framework models several related outcomes that together provide a broader picture of short-term risk.

First, the system predicts **any-event occurrence**, meaning whether at least one relevant conflict event will take place in the target week. These events come from five categories considered in this study:

battles, explosions or remote violence, violence against civilians, air or drone activity, and strategic developments.

Second, the system predicts **high-intensity conflict**, which is intended to capture more severe future weeks. A target week is considered high intensity if it reflects substantial conflict activity, elevated fatalities, or direct violence against civilians.

Third, the framework performs **event-type forecasting**, where it predicts the future occurrence of each event category separately for a given raion and target week. This provides a more detailed view of what kind of escalation may occur, rather than only whether any event will happen.

Finally, the system performs **count regression** to estimate future weekly totals for key conflict outcomes, including event count, fatalities, and air or drone strike count.

Taken together, these tasks allow AccessWatch to move beyond a simple yes-or-no warning system and instead produce a structured forecast of both the likelihood and the form of near-term conflict-related disruption.

4 Dataset Description

4.1 Data sources

To train and evaluate AccessWatch, we combined several datasets (Table 1) that capture complementary aspects of conflict dynamics, disruption signals, and geographic context in Ukraine. These sources were used to construct the multimodal raion-week panel on which the forecasting framework was trained, validated, and tested. Broadly, the dataset integrates conflict-event records, satellite- derived indicators, administrative boundary data, demographic and infrastructure context layers, and historical damage products. After spatial alignment to raions and temporal aggregation to weekly intervals, these sources were merged into a unified forecasting dataset.

4.2 Data collection and preprocessing

All dynamic inputs were aggregated to a common weekly timeline and spatially aligned to 139 raions. Point-based observations were assigned to raions through spatial joins, while polygon and line-based layers were intersected with raion boundaries to derive count-based or area-based summaries. For the UNOSAT-derived static damage features, only records dated before 2024 were retained to avoid leakage of future information.

We used two input configurations in the forecasting experiments. The first, referred to as the *base multimodal input configuration*, includes dynamic features derived from FIRMS and nighttime-light data together with static features describing transport, border access, population, settlement structure, and UNOSAT damage history. The second, referred to as the *lagged-ACLED-enhanced input configuration*, includes all features from the base multimodal configuration and adds ten lagged conflict-history variables. These additional variables consist of lag-1 values and four-week rolling summaries of prior event count, fatalities, air/drone strike count, any-event status, and high-intensity status.

4.3 Temporal coverage, split design, and dataset size

The released manifest reports 104 unique weekly anchors with the following chronology: training from 2023-01-02 to 2024-06-24 (78 weeks), validation from 2024-07-01 to 2024-09-23 (13 weeks), and test anchors beginning 2024-09-30. Because the task is direct forecasting, the usable number of trailing test weeks decreases as the horizon increases.

The test panel sizes decrease from 1,668 rows at week+1 to 1,112 rows at week+5, while the test-set any-event prevalence remains close to 47–48% and the high-intensity prevalence remains close to 21–22%. This stability is useful because it allows fairer cross-horizon performance comparisons.

4.4 Feature construction

The forecasting panel was built from two input configurations. The first is the *base multimodal input configuration*, which combines satellite-derived signals with static geographic and infrastructure context. The second is the *lagged-ACLED-enhanced input configuration*, which contains all features in the base configuration and additionally includes lagged conflict-history variables derived from prior weekly conflict activity. This design allows us to isolate the contribution of recent conflict history beyond the multimodal baseline.

Table 3 summarizes the feature composition of these two input configurations.

4.5 Feature construction

The forecasting panel was built using two input configurations. The first, referred to as the *base multimodal input configuration*, combines satellite-derived signals with static geographic, demographic, infrastructure, and historical damage information. The second, referred to as the *lagged-ACLED-enhanced input configuration*, contains all features from the base configuration and additionally includes lagged conflict-history variables derived from recent weekly conflict activity. This comparison allows us to evaluate how much short-term conflict memory improves forecasting beyond the multimodal baseline.

Table 3 summarizes the feature composition of these two input configurations.

4.6 Construction of the high-intensity label

In addition to the any-event target, we constructed a binary *high-intensity* label to distinguish more severe conflict weeks from lower-level activity. This label was not manually annotated. Instead, it was generated automatically from future-week raion-level aggregates after spatial assignment and weekly temporal aggregation. We adopted this rule-based strategy because the dataset does not provide a direct manually labeled severity indicator at the raion-week level, while a deterministic definition yields a consistent supervision signal for forecasting.

Table 1: Datasets used to train and evaluate AccessWatch.

Dataset name	Description	Data Fields	Data Modality	Year
ACLED conflict-event data	Used to construct both the historical conflict features and the future prediction targets. These records provide the event-level information needed to derive weekly event counts, fatalities, event type-specific indicators, and the high-intensity label.	Event date, event type, sub-event type, fatalities, geographic coordinates, and raion-level spatial assignment.	Structured georeferenced event records (tabular data)	2023–2024
NASA FIRMS thermal anomaly data	Used as a dynamic remote-sensing signal of disruption. Weekly raion-level summaries from FIRMS were included to capture the frequency, intensity, and confidence composition of thermal hotspot activity.	Detection date, geographic coordinates, confidence level, fire radiative power (FRP), brightness measures, day/night flag, and sensor/type indicators.	Satellite-derived point detections	2023–2024
Black Marble nighttime-light data	Used to capture illumination level, short-term change, and observation quality. These features provide a complementary signal that may reflect disruption, instability, or changes in activity patterns.	Weekly mean radiance, median radiance, standard deviation, valid-pixel count, observation-day count, high-quality observation share, lagged change, percentage change, and rolling summaries.	Satellite-derived raster data	2023–2024
Ukraine admin-2 (raion) boundaries	Used as the common spatial framework for all aggregation and modeling. All dynamic and static inputs were aligned to this raion-level geography, and all predictions were generated at the raion-week level.	Raion identifier, raion name, oblast name, and polygon geometry.	Vector polygon boundary data	Static boundary layer
Population, settlement, and infrastructure context layers	Used to provide static contextual information describing exposure, settlement structure, and transportation connectivity for each raion. These variables help characterize baseline population distribution and access-related conditions.	Population totals and density, populated-place counts, road length, major-road length, paved-road share, rail length, bridge-related measures, border-crossing count, and nearest border-crossing distance.	Mixed geospatial data (raster and vector)	Primarily 2022–2023 for population; static reference layers for infrastructure and settlements
UNOSAT damage/disruption products	Used to derive pre-2024 structural damage and disruption features. These variables capture accumulated physical impact, including damage points, damage polygons, flood-related features, affected roads, and affected urban areas.	Damage feature count, damage point count, damage polygon count, damage area, flood feature count, flood area, affected road count, affected urban area, year-specific feature counts, and number of unique source layers.	Vector geospatial damage-assessment layers	Pre-2024, primarily 2022–2023

Table 2: Held-out test-set characteristics by forecast horizon. Positive rates are computed from the true test labels.

Horizon	Rows	Weeks	Raions	Any-event (%)	High-int. (%)
week+1	1668	12	139	47.9	21.5
week+2	1529	11	139	48.0	21.5
week+3	1390	10	139	47.3	21.7
week+4	1251	9	139	47.4	21.7
week+5	1112	8	139	47.4	21.5

Formally, the high-intensity label for the target week at forecast horizon h was defined as

$$y_{t+h}^{(\text{high})} = \mathbb{1}\begin{pmatrix} \text{event\_count}_{t+h} \geq 10 \ \vee \\ \text{fatalities}_{t+h} \geq 5 \ \vee \\ \text{vac\_count}_{t+h} > 0 \end{pmatrix}.$$

Here, event_count_{t+h} denotes the total number of events in the target week, fatalities_{t+h} denotes the total number of reported fatalities in that week, and vac_count_{t+h} denotes the number of *violence-against-civilians* events recorded in that raion during the target week.

This definition was chosen to capture three complementary dimensions of severe conflict conditions: sustained event volume, lethality, and direct violence against civilians. The threshold for event_count, namely ≥ 10, was selected to mark weeks with concentrated conflict activity rather than isolated incidents. In the week+1 training split, the median weekly event count across all raion-weeks is 0 and the 75th percentile is 3, so a cutoff of 10 lies well above routine activity levels and identifies the upper tail of more active

Table 3: Feature composition of the two input configurations used in AccessWatch. The lagged-ACLED-enhanced configuration contains all features in the base multimodal configuration and adds ten lagged conflict-history variables. The final column describes what each feature group represents.

Feature group	Base multimodal input	Lagged-ACLED-enhanced input	What these features represent
Lagged conflict-history features	0	10	Recent conflict dynamics derived from prior weeks, including the previous week's event level and four-week cumulative history of event count, fatalities, air/drone activity, overall event occurrence, and high-intensity status.
FIRMS thermal anomaly features	17	17	Weekly summaries of thermal hotspot activity, including how often hotspots were detected, their confidence levels, their thermal intensity, and how these signals changed relative to earlier weeks.
Nighttime-light features	13	13	Weekly summaries of nighttime illumination, including average brightness, variability, observation quality, and short-term changes that may indicate disruption or recovery.
Settlement-related feature	1	1	The number of populated places within a raion, used as a simple indicator of settlement concentration.
Static context and historical damage features	36	36	Baseline characteristics of each raion, including population size and density, settlement structure, road and rail connectivity, border-access measures, and accumulated pre-2024 damage and disruption recorded by UNOSAT.
Total number of features	**67**	**77**	The enhanced configuration extends the base multimodal input by adding ten lagged conflict-history variables.

weeks. Empirically, only about 16.2% of training raion-weeks satisfy this condition.

The threshold fatalities ≥ 5 was chosen to capture severe weeks that may not have extremely high event counts but are nevertheless highly lethal. In the same training split, the median weekly fatalities count is 0, the 75th percentile is also 0, and the 90th percentile is 4. A cutoff of 5 therefore places the threshold just beyond the 90th percentile of the training distribution, making it a deliberately strict indicator of lethal escalation. Approximately 9.5% of training raion-weeks satisfy this condition.

Finally, the condition vac_count > 0 was included because any violence-against-civilians event is operationally significant in a humanitarian setting. Unlike general event volume, this signal directly reflects harm directed at civilians and may indicate a substantial deterioration in local safety and access conditions even when the total number of events is not especially large. In the training data, such weeks are relatively rare, occurring in approximately 4.6% of raion-weeks.

Taken together, these criteria provide a rule-based severity label that is both substantively meaningful and statistically usable. The resulting positive class is neither too rare nor too broad: in the week+1 training split, the constructed high-intensity label is positive for approximately 20.3% of raion-weeks, and in the held-out test sets its prevalence remains stable at roughly 21–22% across forecast horizons.

4.7 Assumptions and trade-offs

The dataset construction process makes several deliberate trade-offs. Weekly aggregation suppresses daily noise but can smooth short spikes. Raion-level forecasting improves interpretability and tractability but is coarser than point-level modeling. Satellite features are indirect proxies rather than direct access observations, and UNOSAT summaries are static pre-2024 structural indicators rather than continuously updated damage streams.

5 Methodology

5.1 Forecasting setup and prediction tasks

AccessWatch is built on a multimodal raion-week forecasting panel. Rather than operating directly on raw text, raw satellite imagery, or raw geospatial layers, the framework first converts all source data into structured numeric features through spatial aggregation, temporal aggregation, and feature engineering. Each training example, therefore, represents a Ukrainian raion at a specific weekly anchor point using a fixed-length feature vector. For temporal neural models, these weekly feature representations are further arranged into short ordered sequences so that recent temporal patterns can be modeled explicitly.

Given an anchor week, the system predicts outcomes for a target week located exactly h weeks ahead, where $h \in \{1, 2, 3, 4, 5\}$. This direct forecasting design avoids the error accumulation that can occur in recursive forecasting and makes it possible to optimize models separately for each forecast horizon.

The framework addresses four related prediction tasks, summarized in Table 4. The first is any-event prediction, which forecasts whether at least one event will occur in the target week. The second is high-intensity prediction, which forecasts whether the target week will reach a more severe level of conflict activity. The third is event-type prediction, formulated as a multi-label task that predicts the occurrence of each event category separately.

Any-event and high-intensity prediction are binary classification tasks, and event-type prediction is a multi-label classification task.

5.2 Hierarchical output structure

The prediction framework follows a hierarchical output design. At the top level, the model first predicts whether any event will occur in the future target week. This parent prediction then provides the basis for the child tasks, namely high-intensity forecasting, and event-type forecasting.

This structure reflects the underlying logic of the problem. If no event occurs in the target week, the week cannot be classified as high-intensity, and none of the event-type-specific outputs should be positive. Organizing the outputs hierarchically, therefore, helps maintain consistency between broader and more specific forecasts.

In practice, this hierarchy is also incorporated into prediction. The any-event predictor is learned first, and downstream outputs are conditioned on the predicted event probability. As a result, the final framework behaves as a hierarchy-aware forecasting system rather than a collection of unrelated models.

5.3 Input configurations

To evaluate the contribution of different information sources, AccessWatch uses two input configurations. The first is the base multimodal configuration, which includes dynamic features derived from NASA FIRMS thermal anomaly data and VIIRS Black Marble nighttime-light data, together with static features that describe transport infrastructure, border access, population, settlement structure, and pre-2024 UNOSAT damage history.

The second is the lagged-ACLED-enhanced configuration, which contains all features from the base multimodal setup and additionally includes ten lagged conflict-history variables derived from recent weekly conflict activity. These added variables capture lag-1 values and four-week rolling summaries of event count, fatalities, air or drone activity, any-event status, and high-intensity status.

This two-configuration design makes it possible to isolate the value of short-term conflict memory beyond the multimodal baseline. In other words, it allows the study to examine not only whether multimodal data are useful, but also how much additional predictive value comes from explicitly modeling recent conflict history.

5.4 Predictive models

AccessWatch compares five predictive model families under the same forecasting setup. Three of these models operate directly on the engineered raion-week feature vector, while two are designed to model short temporal histories.

The first model serves as a linear baseline. For binary classification tasks, it is implemented as a pipeline with median imputation, feature standardization, and logistic regression. For count-valued regression tasks, the same preprocessing pipeline is followed by ridge regression. This baseline provides a simple linear reference point against which more flexible models can be compared.

The second and third models are LightGBM and CatBoost. Both are gradient-boosted decision-tree ensembles designed for structured tabular data, and both can capture nonlinear relationships and interactions among the engineered features. In this framework, they are used for both classification and regression, providing stronger nonlinear alternatives to the linear baseline while still operating on the same feature representation.

The final two models are temporal neural architectures: GRU and TCN. The GRU-based model uses a two-layer bidirectional gated recurrent unit encoder over recent weekly dynamic features, allowing it to learn short-range temporal dependencies from ordered histories. The TCN-based model uses a temporal convolutional encoder composed of three residual one-dimensional convolution blocks with increasing dilation factors, enabling it to capture temporal patterns across multiple time scales.

In both neural models, dynamic features are organized as weekly sequences, while static features are processed separately through a multilayer perceptron. The temporal and static representations are then fused before prediction. This design allows the neural models to combine short-term temporal structure with persistent geographic and infrastructural context in a unified forecasting framework.

6 Experimental Setup

6.1 Experimental protocol

We evaluate AccessWatch under a direct multi-horizon forecasting setup in which separate targets are constructed for each forecast horizon rather than predicting an aggregated future window. Concretely, for each raion and anchor week t, the system predicts outcomes at a target week located exactly h weeks ahead. The released experiment package evaluates five direct horizons, from week+1 through week+5, and trains a separate model for each horizon. This design avoids recursive error propagation and allows each horizon to be optimized and analyzed independently.

The forecasting panel is split chronologically into training, validation, and test periods. The released data manifest contains 104 weekly anchors, with 78 training weeks, 13 validation weeks, and 13 test weeks before horizon-specific truncation. Because the task is direct forecasting, the number of usable test rows decreases as the horizon becomes longer, since later anchor weeks do not have a complete future target window. All results are therefore reported under strict temporal evaluation rather than random shuffling.

6.2 Model variants and input settings

Experiments are conducted for two input configurations and multiple model families. The first input configuration, *Model 1*, uses the base multimodal feature set. The second, *Model 2*, extends this representation with ten lagged conflict-history variables. The reported benchmark compares five learned model families: a regularized linear baseline, LightGBM, CatBoost, GRU, and TCN. In addition, the training launcher supports an optional naive hierarchical baseline for reference, although the main paper reports the five learned model families.

To test robustness under incomplete multimodal availability, each model-input pair is evaluated under five missing-modality settings: *full*, *drop_firms*, *drop_ntl*, *drop_ntl_firms*, and *drop_unosat*. Combined with five forecast horizons, two feature sets, and five learning algorithms, this yields the full held-out experimental grid described in the report.

Table 4: Prediction tasks used in AccessWatch.

Prediction task	Learning type	Output	Description
Any-event prediction	Binary classification	$\{0, 1\}$	Predicts whether at least one event from any of the five event categories will occur in the future target week for a given raion.
High-intensity prediction	Binary classification with hierarchical dependency	$\{0, 1\}$	Predicts whether the future target week reaches a higher level of severity, conditional on event occurrence.
Event-type prediction	Multi-label classification	$\{0, 1\}^5$	Predicts the occurrence of each event category separately in the future target week: battles, explosions/remote violence, violence against civilians, air/drone activity, and strategic developments.
Count prediction	Regression	$\mathbb{R}$	Predicts future event count, fatalities sum, and air/drone strike count for the target week.

6.3 Training configuration

The direct forecasting experiments are launched through a horizon-specific training script that constructs future targets independently for each requested value of h. The script is parameterized by a random seed of 42 and uses separate validation and test spans of 13 weeks each. For sequence models, the input history length is set to 8 weeks. The default neural training configuration uses 20 epochs, batch size 128, hidden dimension 96, dropout 0.2, learning rate 10^{-3}, weight decay 10^{-4}, and early stopping patience of 5 epochs. Device selection is handled automatically at runtime.

For tabular models, the same horizon-specific split is used, and each model is trained independently for each input configuration and missing-modality condition. For neural models, dynamic features are organized into ordered weekly sequences, while static features are processed separately and fused with the temporal representation before prediction. The training launcher also exposes top-k and top-fraction control parameters, set by default to 10 and 0.2, respectively, for hierarchical prediction and export.

6.4 Target construction and evaluation outputs

For each forecast horizon, future labels are created by sorting observations by raion and week and then shifting the target variables forward by exactly h weeks within each raion. This procedure is applied to the main targets, including any-event occurrence, high-intensity status, and count-valued outcomes, as well as the event-type-specific binary targets. Rows without a valid future any-event label are excluded from the corresponding horizon-specific dataset.

Each experiment produces predictions for the training, validation, and test splits. The exported evaluation files include hierarchy-aware classification metrics for any-event and high-intensity prediction, event-type performance summaries, and regression metrics for count-valued targets. In particular, the summary outputs track any-event F1, high-intensity F1, average precision for the binary tasks, macro/micro/weighted F1 for event-type prediction, and mean absolute error for event count, fatalities, and air/drone strike count. The implementation also records hierarchy diagnostics, such as the fraction of cases in which the predicted high-intensity signal exceeds the predicted event-occurrence signal, to help verify consistency across parent and child outputs.

6.5 Reproducibility and result aggregation

The experiment runner saves per-split prediction exports for every horizon, model setting, and algorithm, along with horizon-level comparison tables and JSON summaries of thresholds and evaluation metrics. After all horizons are completed, the results are aggregated into a single cross-horizon comparison file and a held-out test leaderboard. This design supports reproducible analysis across horizons while also making it easy to identify the strongest configuration for a given task or deployment preference.

7 Experimental Results

7.1 Evaluation grid

The held-out evaluation covers five direct forecast horizons (week+1 through week+5), two feature sets, five predictive model families, and five input-availability settings, yielding 250 test configurations. We report classification quality with F1 and average precision, and count-valued outcomes with mean absolute error (MAE). Because the study compares both full-input and reduced-input settings, we separate two questions in the analysis: which configuration is the strongest *balanced full-input system*, and which configuration attains the *single best score* for a specific horizon-task pair.

7.2 Effect of lagged conflict history under full input

The clearest result in the full-input setting is the benefit of adding lagged ACLED history. Table 5 summarizes this effect by comparing the base multimodal input against the lagged-ACLED-enhanced input under full modality availability.

Across all five model families and all five forecast horizons, adding lagged ACLED history raises average any-event F1 from 0.7947 to 0.8194, high-intensity F1 from 0.7748 to 0.8448, and event-type macro-F1 from 0.6537 to 0.6963. The largest gain appears on high-intensity prediction (+0.0701 F1), which is consistent with the

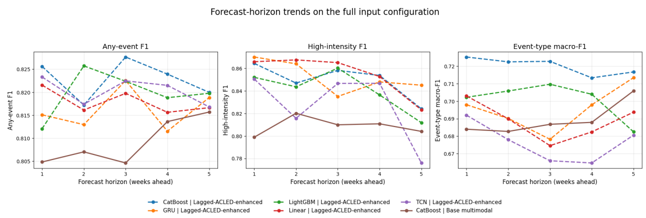

Figure 1: Full-input classification trends across forecast horizons. The lagged-ACLED-enhanced CatBoost model remains the strongest balanced configuration, while the high-intensity task stays comparatively stable even at longer horizons.

intuition that recent conflict history is particularly informative for short-horizon escalation forecasting.

7.3 Full-input model comparison

Under full modality availability, the lagged-ACLED-enhanced input with CatBoost is the strongest *balanced* system. Table 6 shows that, averaged over horizons, this configuration is best on any-event F1 (0.8229) and event-type macro-F1 (0.7201). The lagged linear baseline attains the highest average high-intensity F1 (0.8548), but that gain should be interpreted cautiously: the same model is extremely unstable on the count-regression targets, as shown in Table 7. For that reason, it is not the best overall deployment choice even though it peaks on one classification metric.

Among the non-linear models, LightGBM and CatBoost are clearly the strongest full-input regressors once lagged ACLED history is included. Lagged LightGBM achieves the lowest average event-count MAE (3.8561), fatalities MAE (9.1026), and air/drone-strike MAE (2.3576), while lagged CatBoost remains very close on the same targets and is noticeably stronger on any-event and event-type classification. This is why CatBoost is the more balanced choice when a single model must support the entire hierarchy.

7.4 Week-by-week behavior of the strongest balanced system

Table 8 reports the week-by-week full-input results for the lagged-ACLED-enhanced CatBoost model. Its performance degrades only modestly with forecast distance. Any-event F1 stays between 0.8172 and 0.8276 across week+1 to week+5. High-intensity F1 remains above 0.85 through week+4 and is still 0.8240 at week+5. Event-type macro-F1 also remains stable, ranging from 0.7133 to 0.7252.

The event-type breakdown is also informative. For the full-input lagged CatBoost model, the average event-type F1 across horizons is 0.8285 for battles, 0.7985 for explosions/remote violence, 0.8059 for air/drone activity, 0.6697 for strategic developments, and 0.4977 for violence against civilians. The relative difficulty ordering is therefore quite clear: battles, explosions/remote violence, and air/drone activity are the easiest labels; strategic developments are intermediate; and violence against civilians is the hardest event type by a substantial margin. Figure 2 shows that this ranking is largely preserved across all five horizons rather than being driven by a single week.

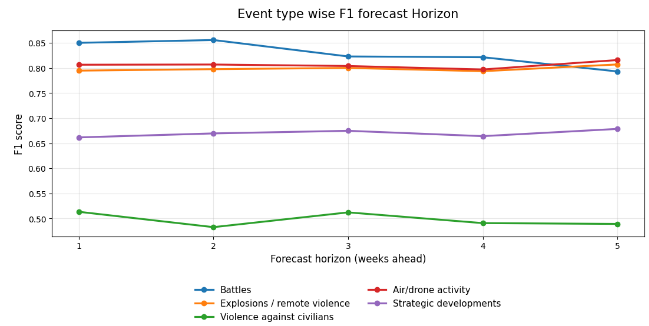

Figure 2: Event-type F1 across forecast horizons for the full-input lagged-ACLED-enhanced CatBoost model. Violence against civilians is consistently the hardest event type, while battles and air/drone activity remain the most predictable.

7.5 Best score versus best balanced system

The strongest balanced model is not always the configuration that yields the single best score for a specific horizon-task pair. Table 9 makes this explicit for the three classification tasks. Two patterns stand out. First, 12 of the 15 best classification cells come from the lagged-ACLED-enhanced feature set. Second, all 15 of the best classification cells come from a reduced-input variant rather than the full-input variant. The full-input system is therefore the most defensible general-purpose setting, but the single best score for a narrow task is often obtained after removing at least one modality block.

The peak high-intensity scores are especially concentrated: GRU attains the best week+1 value (0.8759) with lagged features and no Black Marble nighttime-light input, while TCN attains the best value at week+2 through week+5 under reduced-input lagged configurations. Event-type macro-F1 is dominated by CatBoost under reduced-input settings, which is consistent with the view that event-type forecasting benefits from strong nonlinear tabular modeling but does not necessarily require every modality at weekly raion scale.

8 Ablation Study

We report the missing-modality ablation separately for the two input configurations, since the interpretation changes once lagged conflict-history variables are introduced. This distinction matters. The base multimodal configuration contains engineered weekly features derived from FIRMS thermal anomalies, Black Marble nighttime lights, UNOSAT damage/disruption products, and population, settlement, and infrastructure context layers. The lagged-ACLED-enhanced configuration adds recent conflict-history variables on top of that base representation. As a result, the ablation should be read as a study of *marginal contribution conditioned on the rest of the input set*, not as a universal ranking of modality importance.

8.1 Base multimodal ablation

Table 10 shows the missing-modality ablation for the base multimodal input, averaged over all five model families and all five forecast horizons. In this setting, the strongest negative result comes

Table 5: Effect of adding lagged ACLED conflict-history variables under the full-input setting. Classification averages are computed over all five model families and five forecast horizons. Regression averages are computed over the four non-linear models only, since the linear baseline is poorly calibrated for count-valued targets. Positive values in the delta row indicate higher F1; negative values indicate lower MAE.

Feature set	Any-event F1	High-intensity F1	Event-type macro-F1	Event-count MAE	Fatalities MAE	Air/drone-strike MAE
Base multimodal input	0.7947	0.7748	0.6537	6.3780	12.8470	4.0049
Lagged-ACLED-enhanced input	0.8194	0.8448	0.6963	4.5573	11.4091	2.8676
Δ (Lagged − Base)	+0.0247	+0.0701	+0.0426	−1.8207	−1.4379	−1.1373

Table 6: Full-input classification performance averaged over forecast horizons. Best value in each column is bolded.

	Base multimodal input			Lagged-ACLED-enhanced input		
Model family	Any-event F1	High-intensity F1	Event-type macro-F1	Any-event F1	High-intensity F1	Event-type macro-F1
Linear	0.7451	0.6371	0.5648	0.8179	**0.8548**	0.6887
LightGBM	0.8036	0.7948	0.6791	0.8197	0.8407	0.7008
CatBoost	0.8092	0.8087	0.6894	**0.8229**	0.8493	**0.7201**
GRU	0.8079	0.8245	0.6774	0.8162	0.8523	0.6955
TCN	0.8079	0.8086	0.6578	0.8203	0.8270	0.6763

Table 7: Full-input regression performance averaged over forecast horizons. Best non-linear value in each column is bolded. The linear baseline is retained for reference only.

	Base multimodal input			Lagged-ACLED-enhanced input		
Model family	Event-count MAE	Fatalities MAE	Air/drone-strike MAE	Event-count MAE	Fatalities MAE	Air/drone-strike MAE
LightGBM	5.5041	11.5567	3.7681	**3.8561**	**9.1026**	**2.3576**
CatBoost	5.4836	11.4577	3.5918	3.8630	9.5163	2.3650
GRU	7.0080	14.0655	4.3243	5.6793	13.5812	3.3421
TCN	7.5162	14.3080	4.3356	4.8309	13.4363	3.4058
Linear	11.6802	7208.8724	7200.6764	90.4182	5999.7756	7291.4904

Table 8: Lagged-ACLED-enhanced CatBoost under full modality availability, reported separately for week+1 through week+5.

Forecast horizon	Any-event F1	High-intensity F1	Event-type macro-F1
week+1	0.8256	0.8643	0.7252
week+2	0.8172	0.8468	0.7224
week+3	0.8276	0.8581	0.7227
week+4	0.8239	0.8536	0.7133
week+5	0.8200	0.8240	0.7167

from removing UNOSAT. Relative to the full base input, dropping UNOSAT reduces any-event F1 from 0.7947 to 0.7915, high-intensity F1 from 0.7748 to 0.7536, and event-type macro-F1 from 0.6537 to 0.6500. The reduction in high-intensity prediction is especially notable (−0.0212 F1), and is the largest classification drop among the base-feature ablations.

This pattern is consistent with the regression results as well. When UNOSAT is removed from the base multimodal input, event-count MAE increases from 6.3780 to 6.5363, fatalities MAE increases from 12.8470 to 13.0141, and air/drone-strike MAE increases from 4.0049 to 4.2356. Taken together, these results indicate that UNOSAT damage/disruption products provide useful structural context when recent conflict history is *not* explicitly available to the model.

The other modality removals are more mixed. Removing both Black Marble nighttime-light and FIRMS features yields the highest average any-event F1 (0.8009) and the highest event-type macro-F1 (0.6859), while removing nighttime-light alone gives the highest average high-intensity F1 (0.7788). These gains are real, but they should not be over-interpreted: they indicate that some aggregated remote-sensing signals are noisy or partially redundant at weekly raion scale, not that the entire multimodal block is unnecessary. In particular, the UNOSAT results show the opposite.

9 Conclusion

AccessWatch demonstrates that a hackathon-origin system can still support serious multimodal forecasting analysis when it is built around disciplined panel construction, strict temporal evaluation, and hierarchy-aware prediction. The released package covers 139 raions, 104 weekly anchors, five direct forecast horizons, five algorithms, and five missing-modality settings. The best balanced full-input system is Model 2 with CatBoost, which provides strong event and high-intensity classification together with stable event type and count performance.

Table 9: Best classification configuration at each forecast horizon across all 250 held-out test configurations.

Forecast horizon	Best any-event F1	Best high-intensity F1	Best event-type macro-F1
week+1	GRU (Lagged-ACLED-enhanced, No NASA FIRMS): 0.8298	GRU (Lagged-ACLED-enhanced, No Black Marble NTL): 0.8759	CatBoost (Lagged-ACLED-enhanced, No NTL + FIRMS): 0.7351
week+2	TCN (Base multimodal, No NTL + FIRMS): 0.8282	TCN (Lagged-ACLED-enhanced, No NASA FIRMS): 0.8739	CatBoost (Base multimodal, No NTL + FIRMS): 0.7236
week+3	GRU (Lagged-ACLED-enhanced, No NTL + FIRMS): 0.8333	TCN (Lagged-ACLED-enhanced, No NASA FIRMS): 0.8784	CatBoost (Lagged-ACLED-enhanced, No NTL + FIRMS): 0.7295
week+4	TCN (Lagged-ACLED-enhanced, No NTL + FIRMS): 0.8284	TCN (Lagged-ACLED-enhanced, No NTL + FIRMS): 0.8627	CatBoost (Lagged-ACLED-enhanced, No UNOSAT): 0.7272
week+5	TCN (Lagged-ACLED-enhanced, No UNOSAT): 0.8348	TCN (Lagged-ACLED-enhanced, No NTL + FIRMS): 0.8606	TCN (Base multimodal, No NTL + FIRMS): 0.7263

Table 10: Missing-modality ablation for the base multimodal feature set, averaged over all five model families and all five forecast horizons. Positive deltas indicate higher F1 than the full-input condition.

	Any-event		High-intensity		Event-type macro	
Input configuration	F1	Δ vs full	F1	Δ vs full	F1	Δ vs full
Full input	0.7947	0.0000	0.7748	0.0000	0.6537	0.0000
No Black Marble NTL	0.7941	−0.0007	**0.7788**	**+0.0040**	0.6598	+0.0061
No NASA FIRMS	0.7996	+0.0049	0.7643	−0.0104	0.6619	+0.0082
No NTL + FIRMS	**0.8009**	**+0.0061**	0.7677	−0.0070	**0.6859**	**+0.0322**
No UNOSAT	0.7915	−0.0033	0.7536	−0.0212	0.6500	−0.0037

The results also show that more modalities do not automatically imply better predictions. The most important gain comes from adding autoregressive conflict-history features, while the marginal contribution of FIRMS and nighttime-light inputs depends on the learner and horizon. This is a scientifically meaningful result because it shows that multimodal forecasting quality depends on signal complementarity, not only on modality count.

Future work should extend the system with more frequently refreshed infrastructure-damage layers, better calibrated uncertainty estimates, and access-ground-truth outcomes that go beyond conflict proxies alone. Even in its current form, however, AccessWatch provides a strong and publishable foundation for humanitarian-access early warning research.